

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Digital Logic

Semester: Spring

Year : 2026
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Convert the following conversions. 8
- i. $(215.AC)_{16} = (?)_8$
 - ii. $(110111101)_2 = (?)_{\text{gray}}$
 - iii. $(101.11)_2 - (1010.11)_2$ using 2's complement
 - iv. If $(150)_8 = (65)_x$, find the value of x .

- b) Discuss on the University of NAND and NOR gates. 7

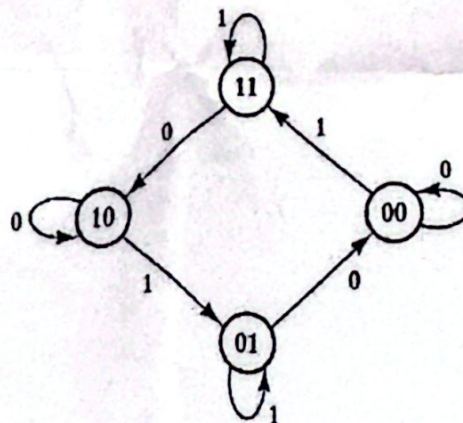
2. a) Draw K-map in SOP form for expression $F = \pi M(1,3,8,10,12,15)$ and $D = \sum (0,7,11,14)$? Also realize using Two Level NOR Gates only. 7

OR

Using K-map, simplify the Boolean function

$F(A,B,C,D) = \sum(0,2,3,5,7,8,10,11,15)$ with don't care condition $d(A,B,C,D) = \sum(1,6,9)$ in both SOP and POS form.

- b) Design a circuit for parity generator and parity checker for 3 bit message with even parity. 8
3. a) Draw the sequential circuit from the information provided in the finite state automata in the figure below using T flip-flops. 8



- b) What is a priority encoder? Design 3×8 decoder using two 2×4 decoders. 7
4. a) Differentiate between a flip-flop and a latch. Explain the operation of a clocked D flip-flop with its logic diagram, characteristic table, characteristic equation and excitation table. 7
- b) Discuss in detail the master slave flip flop (Use JK-flip). 8
- a) Use T flip-flop to design a 3-bit Synchronous UP Counter and draw the circuit diagram. 8
- b) Explain Ring counter. 7

OR

Define register and discuss on its type.

6. a) Design a 4-bit Arithmetic and Logic Circuit that performs eight different arithmetic operations and four logical operations. 8
- b) Explain the operation of shifter and status register. 7
7. Write short notes on: (Any two) 2×5
- a) Compare and contrast analog and digital system
- b) Primary coding systems and their specific uses
- c) Combinational vs sequential